Research Methodology

Research Problems

Equipping You with Research Depth and Industry Skills

By:

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What is Research?

- **Definition (Simple):**

  Research is a **systematic investigation** to discover new knowledge, validate facts, or solve a problem.

- **Formal Definitions:**
- **Clifford Woody (1927):**

  "Research comprises defining and redefining problems, formulating hypotheses, collecting, organizing, and evaluating data, making deductions, and reaching conclusions."

- **Creswell (2014):**

  "Research is a process of steps used to collect and analyze information to increase our understanding of a topic or issue."

Key Features of Research:

- **Systematic:** follows a structured process.
- **Objective:** unbiased, based on evidence.
- **Original:** adds new knowledge or insight.
- **Replicable:** can be repeated by others.
- **Problem-oriented:** aims to answer a research question.
- **Example (AI Context):**
- Problem: “Can deep learning improve early detection of liver tumors?”
- Research = systematically designing experiments, collecting data, analyzing results → to answer the question.

What is a Research Problem?

- A **specific issue, gap, or question** in existing knowledge that needs investigation.
- Should be:
  - Clear & focused.
  - Researchable (data available).
  - Significant (adds value).

Sources of Research Problems

- **Literature Review** → finding gaps in existing studies.
- **Practical Issues** → problems faced in industry/society.
- **New Trends/Technology** → AI, blockchain, big data, etc.
- **Supervisor's Guidance** → ongoing projects and labs.
- **Your Curiosity** → questions that interest you.

Characteristics of a Good Research Problem

- Clear & well-defined.
- Original (not already solved fully).
- Significant (benefits academia/industry/society).
- Feasible (time, resources, expertise available).
- Researchable (data, methods, tools exist).
- Ethical (no harm or plagiarism).

How to Select a Research Problem

- **Identify a broad area of interest.**
  - Example: Natural Language Processing.
- **Review literature** to spot gaps.
  - Example: Humor detection in Arabic text is underexplored.
- **Narrow down scope.**
  - From “NLP” → “Arabic humor detection.”
- **Evaluate feasibility.**
  - Data availability? Methods/tools? Timeframe?
- **Check relevance.**
  - Does it contribute to knowledge or solve a real problem?

Common Mistakes to Avoid

- Too broad or too vague (impossible to finish).
- Already solved many times (not original).
- No available data/resources.
- Overly ambitious for MS-level research.
- Ignoring ethical issues.

Example Exercise

- **Broad Area:** Cybersecurity in Social Media
- Research Gap: Limited work on **Arabic-language hate speech detection**.
- Feasible Scope: Using **deep learning models (e.g., AraBERT)**.
- Final Research Problem: *“Developing a context-aware deep learning model for detecting hate speech in Arabic social media posts.”*

Example Exercise

- **Broad Area:** Cybersecurity in Social Media
- Research Gap: Limited work on **Arabic-language hate speech detection**.
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Research Gap vs. Research Problem

- **Research Gap**
- A **missing piece** in existing knowledge.
- Something that has not been studied enough or at all.
- Found during the **Literature Review**.
- Example: *“Most studies on cyberbullying detection are in English, very few in Arabic.”*
- **Research Problem**
- The **specific issue/question** you decide to investigate.
- A gap **converted into a researchable problem**.
- Must be clear, feasible, and significant.
- Example: *“How can deep learning models be used to detect cyberbullying in Arabic social media posts?”*

How to Formulate a Research Problem

- **Step 1 – Start with a Broad Topic**
- Choose an area you are interested in.
- Example: *Natural Language Processing (NLP).*
- **Step 2 – Do a Literature Review**
- Find **what has already been done**.
- Identify **gaps/limitations**.
- Example: *Most sentiment analysis research is in English.*
- **Step 3 – Define the Research Gap**
- State **what is missing**.
- Example: *Few studies on sentiment analysis for low-resource languages.*
- **Step 4 – Narrow It Down**
- Focus on one **specific aspect** (method, dataset, application).
- Example: *Arabic sentiment analysis using deep learning.*
- **Step 5 – Frame the Problem Clearly**
- Write it as a **clear, specific, researchable statement**.
- Example:
  "There is limited research on applying deep learning methods for sentiment analysis in Arabic social media text, which hinders effective understanding of public opinion in Arabic-speaking regions."
- **Step 6 – Convert into Research Questions**
- RQ1: *How effective are deep learning models for Arabic sentiment analysis?*
- RQ2: *Which model performs best on Arabic social media datasets?*

Thinking

- How to identify gaps/limitations after doing a Literature Review (LR)

How to Identify Research Gaps & Limitations

- **1. Look for what authors say**
- Check the **“Future Work”** or **“Limitations”** section of papers.
- Authors often admit:
  - Small datasets
  - Narrow scope (e.g., only English)
  - Weak methods
  - Context-specific results
- **Compare Studies**
- Do multiple papers agree or conflict?
  - If results are inconsistent → gap.
- Example: *Paper A says CNNs work best, Paper B says LSTMs work best → unclear → research gap.*
- **3. Check the Scope**
- Are there **languages, regions, datasets, or applications** missing?
- Example: *Many papers on English sentiment analysis, but little on Arabic/Urdu.*
- **4. Methodology Weakness**
- Did past studies use **outdated models**?
- Could **newer methods (e.g., transformers)** do better?
- **5. Time Factor (Recency)**
- If most studies are **5+ years old**, the field may need updates with new data/methods.
- **6. Your Observation**
- Sometimes, your **curiosity** finds gaps others missed.
- Example: *No one studied humor detection in Arabic memes yet.*

Key Message

- **First:** *decide what field excites you* (AI, NLP, Cybersecurity, etc.).
- **Then:** *do a Literature Review (LR)* to see **what's done, what's missing, and where the gap is**.
- **Finally:** *formulate your research problem/question.*